HDOC Day-47

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Aim

To write a C program that finds the index of the smallest element in a sorted variable-length integer array that is greater than or equal to a given target, using pointers.

C Program

```
#include <stdio.h>

int ceil_index(int *arr, int n, int target)
{
    int i;

    for (i = 0; i < n; i++)
    {
        if (*(arr + i) >= target)
        {
            return i;
        }
    }

    return -1;
}

int main()
{
    int n, i, target, index;

    printf("Enter size of array: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter sorted array elements:\n");
    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter target: ");
    scanf("%d", &target);

    index = ceil_index(arr, n, target);

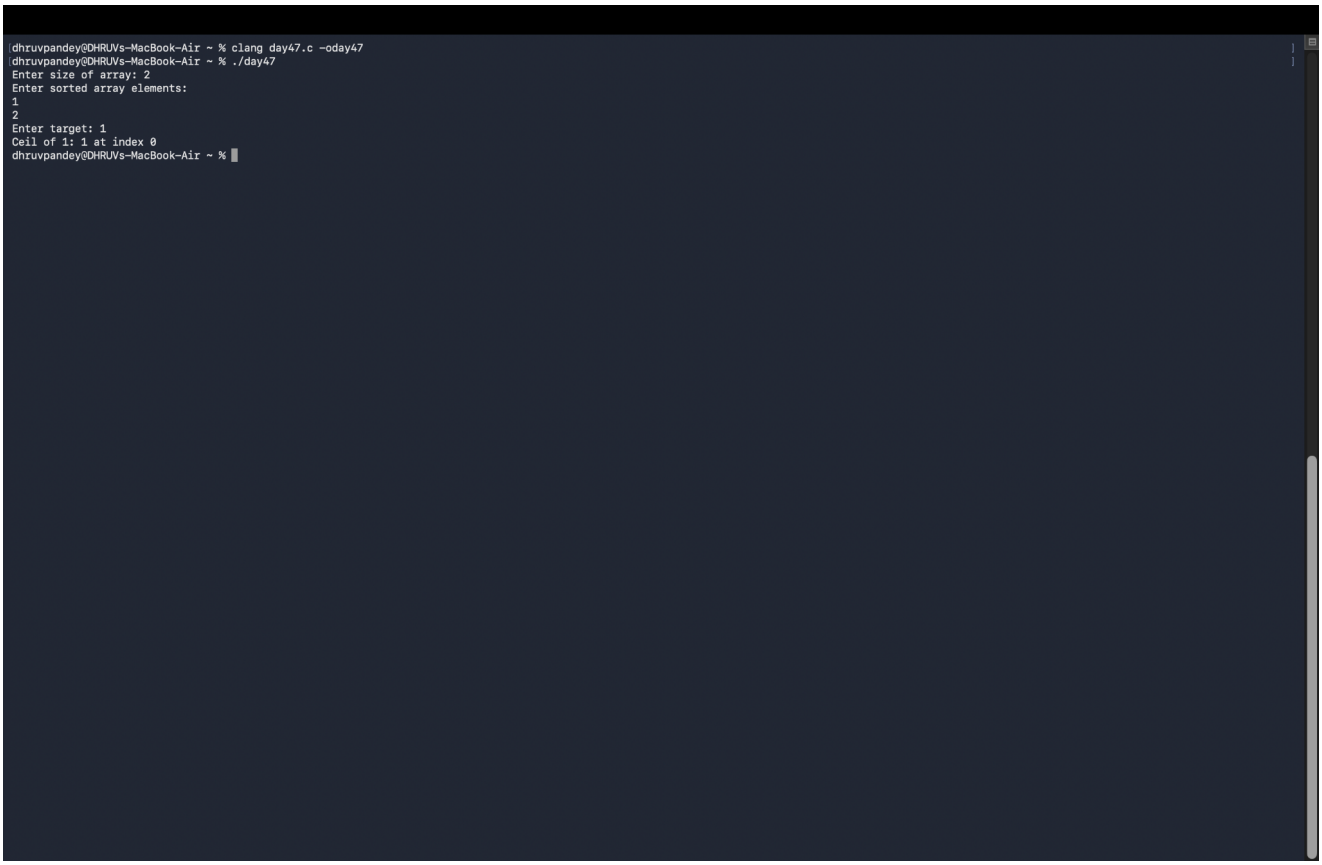
    if (index == -1)
        printf("-1\n");
    else
        printf("Ceil of %d: %d at index %d\n",
               target, arr[index], index);

    return 0;
}
```

Logic

Since the array is sorted in ascending order, the function checks from the beginning. The first element greater than or equal to the target is the ceil. Returning immediately also gives the first occurrence when duplicates are present. If no such element exists, the function returns -1.

Execution


dhruvpandey@DHRUVs-MacBook-Air ~ % clang day47.c -oday47
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day47
Enter size of array: 2
Enter sorted array elements:
1
2
Enter target: 1
Ceil of 1: 1 at index 0
dhruvpandey@DHRUVs-MacBook-Air ~ %